

Flipkart Customer Satisfaction (CSAT) Prediction using Machine Learning

Abstract

This project focuses on analyzing customer support ticket data and developing a machine learning model to classify support requests. The objective is to automate ticket categorization, reduce manual effort, and improve response time. The project includes data preprocessing, exploratory data analysis (EDA), feature engineering, model training, hyperparameter tuning, and performance evaluation.

1. Introduction

Customer support teams receive thousands of support requests daily. Manually categorizing these tickets is time-consuming and prone to errors. Machine learning provides an automated solution by learning patterns from historical support data and accurately predicting ticket categories.

The aim of this project is to build an efficient classification model using customer support data.

2. Problem Statement

Develop a machine learning model capable of accurately classifying customer support tickets based on the available customer and ticket information.

3. Objectives

- Understand the customer support dataset.
 - Clean and preprocess the data.
 - Perform exploratory data analysis.
 - Train multiple machine learning models.
 - Optimize model performance.
 - Evaluate the models using appropriate metrics.
 - Select the best-performing model.
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4. Dataset Description

The dataset contains customer support ticket information with features such as:

- Customer Information
- Ticket Details
- Product Details
- Support Category
- Priority
- Resolution Status

5. Data Preprocessing

The following preprocessing steps were performed:

- Loaded dataset using Pandas.
- Checked missing values.
- Removed duplicate records.
- Handled null values.
- Encoded categorical variables.
- Performed feature selection.
- Split the dataset into training and testing sets.

These preprocessing steps improved the quality of data for model training.

6. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset.

The analysis included:

- Distribution of ticket categories
- Missing value analysis
- Correlation analysis
- Feature distribution
- Count plots
- Heatmap
- Histogram visualization

These visualizations helped identify important features and understand relationships between variables.

7. Feature Engineering

Feature engineering involved:

- Encoding categorical variables
 - Selecting useful features
 - Removing unnecessary columns
 - Preparing data for machine learning algorithms
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8. Machine Learning Models

Multiple machine learning algorithms were implemented and compared.

The models include:

- Decision Tree
 - Random Forest
 - Logistic Regression
 - Support Vector Machine (if implemented)
 - Other models available in the notebook
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9. Hyperparameter Tuning

Hyperparameter tuning was performed to improve model performance.

Techniques used include:

- Grid Search
- Cross Validation

The optimized parameters helped increase model accuracy and reduce overfitting.

10. Model Evaluation

The trained models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report

The evaluation metrics indicate the effectiveness of each model in predicting customer support ticket categories.

11. Results

After comparing all machine learning models, the best-performing model achieved the highest prediction accuracy on the testing dataset.

The optimized model demonstrated improved classification performance and better generalization.

12. Conclusion

This project successfully developed a machine learning model for customer support ticket classification. Proper preprocessing, feature engineering, and hyperparameter tuning significantly improved prediction performance. The developed system can help organizations automate ticket classification, reduce response time, and improve customer satisfaction.

13. Future Scope

Future improvements may include:

- Deep Learning models
- NLP-based ticket analysis
- Real-time ticket classification
- Model deployment using Flask/FastAPI
- Integration with customer support platforms